AJITA SHREE

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GitHub, LinkedIn

EDUCATIONAL QUALIFICATION

Year	Degree	Institute
2020 - present	PhD, Computer Science & Engineering	IIT Kanpur
2015 - 2017	M. Tech, Computer Science & Engineering	IIT Delhi
2010 - 2014	B. Tech, Computer Science & Engineering	MDU Rohtak

ACADEMIC EXPERIENCE

Doctoral Thesis

Supervisor: Prof. Hamim Zafar
Research Focus: Spatial Transcriptomics, Single cell transcriptomics, Machine Learning, Deep generative modelling

- Developing unsupervised learning algorithms for the joint analysis of Single cell & Spatial transcriptomics.
- scDREAMER: atlas-level integration of single-cell datasets using deep generative model paired with adversarial classifier. scDREAMER-sup: supervised integration of single-cell datasets using the hierarchical deep-generative model. *Biorxiv link*
- Single-cell sequencing analysis for mice pancreatic islet in Type-2 diabetes (in collaboration with BSBE dept.). Comparative study of sc-RNA data from regular chow diet mice vs High-fat diet mice to decipher the physiological changes in islet cells under Type 2 Diabetes progression.

Master's Thesis

Topic: Approximation Algorithms for Facility Location with uniform service installation costs
Supervisor: Prof. Naveen Garg
Designed local-search heuristics-based algorithms with improved approximation ratio by a factor of 2, for a one-level hierarchical facility location problem. Improved the worst-case analysis of local search-based algorithms for capacitated version by a factor of 2, under uniform facility costs assumption. Experimental analysis of designed algorithms for the variants showed performance within (1+0.2) factor of optimal solutions.

INDUSTRIAL EXPERIENCE

Data Scientist, GE Aviation, JFWTC, Feb'18 - Aug'20

- **GEnx Flight scheduling:** Developed large-scale probabilistic machine learning systems for engine failure forecasts for the international airlines industry. Collaborated with cross-functional teams across the globe to extend the work across multiple engine lines. The work significantly brought the number of unplanned engine removals and is contributing to GE Aviation's Intellectual Property.
- **On-time Delivery forecasting:** Developed algorithms to predict on-time delivery of engine parts (with its drivers) and devised a way for optimal inventory planning for engine C-parts (nuts, bolts) supply chain.

Research Engineer, GE Global Research, JFWTC, Jul'17 - Jan'18

- Developed supervised models to predict a group of 10 thermo-physical properties of super-alloys towards 2X faster materials discovery suitable for 3D printing. The models are based on custom physics-based objective functions that were solved using global optimization and genetic algorithms.

Software Engineer, CMC Limited, Jan'14 – Jul'14

Developed android applications for locating essential service providers in nearby locations.

TEACHING ASSISTANCE EXPERIENCE

- Data Structures and Algorithms, Inst: Prof. Amitabha Bagchi, ***IIT Delhi (Fall 15)***
- Fundamentals of Computing, ***IIT Delhi* (*Spring 16, Fall 16, Spring 17*), *IIT Kanpur (Fall 20, Spring 22)***
- Statistical Natural Language Processing, Inst: Prof. Ashutosh Modi, ***IIT Kanpur (Spring 21)***
- Deep Reinforcement Learning, Inst: Prof. Ashutosh Modi, ***IIT Kanpur (Fall 21)***
- Head TA, Computational Genomics***,*** Inst: Prof. Hamim Zafar ***IIT Kanpur (Fall 22)***
- Probabilistic Machine Learning, Inst: Prof. Piyush Rai, ***IIT Kanpur (Spring 23)***

CONFERENCES

- *"scDREAMER: atlas-level integration of single-cell datasets using deep generative model paired with adversarial classifier", Poster Presentation, Single Cell Biology 2022, ISMB 2022*

AWARDS & ACHIEVEMENTS

- **Digital Recognition Award – Innovation, *Aug' 20*:** Awarded globally for contributions to GE Aviation.
- **Digital Technology (DT) Leadership Program, GE, Jul'17 - Jul'20:** Graduated from 2-year GE leadership program. During my tenure, collaborated on projects in multi-disciplinary domains.
- **DT eXperience (DTX) - 1, GE Detroit Hub, United States, Jul'18:** Graduated from 2-year global leadership program. DTX-1 was 2 weeks of integrated group training focused on building the fundamentals of leadership through seminars from esteemed leaders.
- **DTX - 2, GE Budapest, Hungary, Aug'18:** Elaborate training on advanced leadership skills with ~100 participants from across the world at GE's Europe headquarter. The training involved visits to multiple GE research labs and manufacturing units in Hungary.
- **GATE-CS, *Mar'2015*:** Secured All India Rank 77 in Computer Science and Engineering.
- **Haryana State M. Vishvesvarya Scholarship, *Aug' 2010*:** Awarded a scholarship of INR 80,000 from Haryana State Government for being the department rank opener.

ACADEMIC PROJECTS & GRADUATE COURSES

- **Tuberculosis Drug Resistance Prediction, Computations Genomics, Inst: Prof. Hamim Zafar, IIT Kanpur (Fall 20):** Developed multi-task classification model for predicting the resistance w.r.t. 11 different drugs.
- **Word sense Disambiguation, Statistical Natural Language Processing, Inst: Prof. Ashutosh Modi, IIT Kanpur (Fall 20):** Developed M-BERT based framework for word-in-context disambiguation for multi-lingual & cross-lingual data. Data included English, French, Spanish, Russian, Chinese & Arabic languages.
- **Optical Character Recognition, *Probabilistic Graphical models, Inst: Prof. Parag Singla,* IIT Delhi (Fall 16):** Designed Bayesian and Markov networks for OCR. Implemented inference algorithms e.g., message passing and Gibbs Sampling to perform inference. Also, performed parameter's learning on the graphical model for future predictions.
- **Pastry Distributed Hash Table (DHT), Software Systems Lab, Inst: Prof. Huzur Saran, IIT Delhi (Fall 15):** Implemented peer-to-peer file sharing system using pastry algorithm along with an application to store and receive information securely.
- **Discrete Event Simulator, Advanced Data Structures, Inst: Prof. S. N. Maheshwari, IIT Delhi (Fall 15):** Developed a simulator to model crowds in an uneven traffic and stampede.
- **ARM-Simulator, Software Systems Lab, Inst: Prof. Huzur Saran, IIT Delhi (Fall 15):** Developed ARM simulator to model the execution of instructions with 5-stage pipeline.
- **Additional Courses:** Topics in Probabilistic modelling and Inference, Machine Learning, Inst: Prof. Piyush Rai, IIT Kanpur (Spring 21, Fall 20); Convex Optimization, Inst: Prof. Amit Kumar, Prof. Nisheeth Vishnoi, IIT Delhi (Spring 17); Data Mining, Database systems, Inst: Prof. Shyam Kumar Gupta, IIT Delhi (Fall 16, Spring 16); Intro. to Machine Learning, Inst: Prof. Parag Singla, IIT Delhi (Spring 16)